When the Grass Is Greener

Three-Shot Search on an Exponentiated Ornstein–Uhlenbeck Landscape

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Abstract

A searcher evaluates a rugged landscape, the exponential of a stationary Ornstein–Uhlenbeck path, at three points chosen sequentially, and is paid the value at the final point: a terminal placement problem, distinct from approximating the path's maximum. The two-evaluation problem has a three-phase closed form: flee below the median, stay above a threshold, otherwise step a distance set by the mean-reversion scale. The third evaluation turns on a bracket of two observed locations and rests on one identity: every interior bracket point shares its conditional mean with an explicit exterior point while carrying conditional variance larger by the constant factor $(1+\rho)/(1-\rho)$, so matched-mean dominance extends to every convex payoff by Gaussian convex ordering. The resulting phase diagram is exact: the interior beats every alternative, observed locations included, on a stated interval of the observed value b, and above an explicit ceiling the grass is not greener, since mean reversion caps the conditional mean of every unvisited point and no variance bonus compensates. Numerically the interior option is worth about one percent of expected value at most, and most of the three-shot bracket widening is explained by the extra evaluation itself rather than by the interior option.

1 Introduction

How many pivots does a firm get? In randomized samples of startups, roughly three quarters pivot zero times over a year of observation, a fifth pivot exactly once, and only six to eleven percent more than once (Camuffo et al. 2020, 2024). The same trials find that teaching founders to search more scientifically shifts them toward exactly one pivot rather than none or many. Strategic search, as practiced, is a game of two or three evaluations.

Callander modeled such search as trial and error on an unknown rugged mapping from strategies to outcomes, drawn as the realized path of a stochastic process: each trial is a strategy actually enacted, and the last one is lived with. The workhorse of the tradition that followed is myopic, each step maximizing the current outcome, and the survey of Bardhi and Callander lists farsighted search with unrestricted direction among its open methodological questions.

This paper solves the smallest farsighted case exactly, on a landscape adapted to strategic search in two ways. Performance is the exponential of a stationary Ornstein–Uhlenbeck path. Mean reversion says that far from anything tried, performance regresses to the industry average, so news is good or bad relative to that average. The exponential says a strategy that truly hits the sweet spot outsells its near neighbors by a wide margin, so residual uncertainty about an untried strategy is valuable in itself.

The searcher holds an incumbent whose performance is known, may try one more strategy, and then must commit. With two evaluations the optimal rule is pivot-or-persevere with exact thresholds: abandon a below-average position for fresh ground, keep an exceptional one, and otherwise step a distance set by how fast the landscape decorrelates. With three, the interesting question is the final move, and the answer is a phase diagram. On bad news, leave past the better of the two positions tried, in whichever direction that lies. On good news, commit to a strategy between them. On news good enough, stay put. The between region has structure of its own: moderate good news sends the searcher to the midpoint, very good news to a blend hugging the better-known position, and the frontier of every regime is closed form.

Two further results cut against the pivot's importance. The option to settle between past positions adds about one percent of expected value at the optimum, so the last pivot's worth lies mostly in the information it gathers, not in the blending it enables. And on a landscape that decorrelates quickly relative to the strategy space, the formulas themselves demote the between option to a second-order refinement: bold fresh probes dominate. Both points are visible in data. On 77,040 radio measurements across a university floor, a physical search field with the same mathematical shape, detrended log performance is Gaussian with exponential spatial correlation ($R^2 = 0.98$), the form radio engineering has used since Gudmundson (1991), and the leading policies in a replay on those measurements are the far-probing ones this regime predicts (Section 10).

2 Related work

Callander (2008) introduced the Brownian policy landscape and its first exact threshold: delegation to a biased expert survives only when the bias is below $\sigma^2/2\mu$. The two 2011 papers made the landscape a search field. A myopic sequence of policymakers or entrepreneurs experiments exactly when the incumbent outcome falls outside $[-\alpha, \alpha]$ with $\alpha = \sigma^2/2|\mu|$, search passes through monotonic and triangulating phases, and it sticks, often at bad outcomes, after one to four placements in simulation (Callander 2011a,b) — an in-model pivot count consistent with the field evidence of the introduction. The line that followed keeps the same shape: exact thresholds in the dimensionless ratio $\sigma^2/|\mu|$ govern precedent selection (Callander and Clark 2017), agenda control (Callander and McCarty 2024), and the performance trap of risk-averse learners (Callander and Matouschek 2019), with the number of evaluations always endogenous and payoffs always flow. The game here differs on the three margins that line never varies: the shot budget is fixed, the payoff is terminal, and the landscape mean-reverts.

Two papers come closest to a fixed-shot phase diagram. Callander and Hummel (2014) analyze a two-shot sequential game between polarized policymakers and obtain a double-cutoff equilibrium ($\sigma^2/4\mu < \gamma^* < \gamma^{**}$) in which the first mover experiments preemptively from a status quo that already delivers his ideal; the shots are strategic and the payoffs flow. Ganz (2020) is the closest decision problem: exactly two shots, one offline trial and one online placement, with terminal-only payoff on a Callander landscape, wrapped in a manager-retention game with partly computational boundaries. The present paper is the three-shot, single-agent member of this progression, solved exactly.

Farsighted search over correlated alternatives has otherwise advanced by restricting something else. Urgun and Yariv (2025) restrict direction: a searcher expands contiguous territory, choosing speed, and stops at a closed-form drawdown; Cetemen, Urgun and Yariv (2023) extend the drawdown analysis to teams, and Wong (2025) to flow payoffs. Bardhi (2024) restricts timing: a batch of k attribute samples chosen at once for an estimation objective, with the placements exact. Garfagnini and Strulovici (2016), Carnehl and Schneider (2025), and Callander, Lambert and Matouschek (forthcoming) study infinite sequences of short-lived experimenters. Aybas and Callander (2025) extend the landscape itself to Ito diffusions, including the Ornstein–Uhlenbeck case, for a one-shot cheap-talk game; to our knowledge that is the only appearance of a mean-reverting landscape in this tradition, and multi-shot search on one appears here first. Bardhi's nearest-attribute theorem supplies the license for the choice: within the powered-exponential family, the OU kernel is the unique covariance satisfying the Markov property that the tradition's updating relies on.

The optimization literature treats the same object with many evaluations and asymptotic goals: Calvin derives average-case rates under Wiener measure, Grill, Valko and Munos approximate the maximum of a Brownian path, and Grosse, Zhang and Hennig extend the approach to general Gaussian-process samples with regret rates in a growing budget. Alabert and Caballero (2018) compute exact laws of a conditioned bridge minimum to compare nonadaptive few-point rules. Malladi (2025) and Banchio and Malladi (2025) obtain closed-form few-shot policies in a worst-case, prior-free setting, where roughness plays the role volatility plays here. Hodgson and Lewis (2025) estimate Gaussian-process search on clickstream data, solving the farsighted problem numerically. None of these solves a fixed small budget of adaptive evaluations on a Bayesian landscape in closed form.

The phase boundaries are behaviorally meaningful as well as derivable: in the laboratory, subjects placed on exactly this landscape respond to Brownian uncertainty but over-modify when complexity is high (Glick and Myers 2015).

3 The model: three evaluations of an exponentiated OU path

Let X be a stationary Ornstein–Uhlenbeck process on $\mathbb{R}$ with $\mathbb{E}[X_t] = 0$, unit marginal variance, and covariance kernel

$$k(s,t) \;=\; e^{-\kappa|s-t|}. \tag{1}$$

The landscape is $f(t) = \exp(X_t)$: locally continuous, mean-reverting in its logarithm, with occasional high peaks where excursions are magnified by the exponential. A searcher samples f at a first location, which we take to be $t_0 = 0$ by translation invariance, discovering $X_0 = b$. Two further evaluations at locations t_1 and then t_2 follow, each chosen with knowledge of all values so far, and the payoff is the value at the final location, $u = \mathbb{E}\,[f(t_2)]$. Conditional on the observations gathered by the time t_2 is chosen, X_{t_2} is Gaussian with some mean μ and variance ν, and

$$\log \mathbb{E}\big[e^{X_{t_2}} \,\big|\, \mathcal{F}\big] \;=\; \mu + \tfrac{1}{2}\nu, \tag{2}$$

where $\mathcal{F}$ denotes the observed locations and values. (The unconditional distribution after adaptive placement need not be Gaussian; every use of (2) below is conditional.) The searcher is thus paid for conditional mean and conditional variance in equal measure: a final location with residual uncertainty carries a lognormal bonus of half its variance.

Figure 1 shows the decision this paper solves. Two conventions tidy the boundary cases. A searcher may take a location to infinity, which by mean reversion is equivalent to sampling

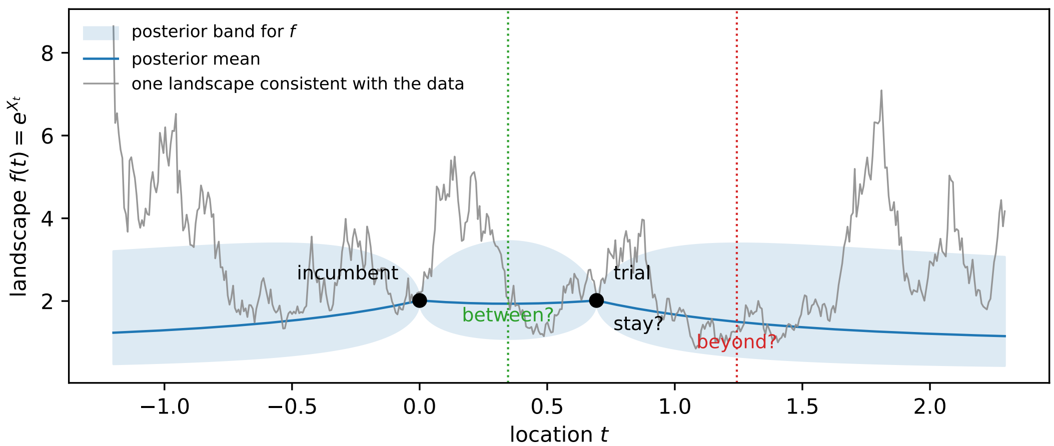

Figure 1: The problem. Two locations have been evaluated and returned equal values (dots); the shaded region is the posterior band for the landscape and the grey curve one landscape consistent with the data. The final evaluation is the consequential one: settle between the anchors, step out past an end, or stay at an observed point. The paper computes which, exactly, as a function of the anchor value and the bracket correlation.

a fresh $N(0,1)$ value independent of everything seen; and a searcher may finish at an observed location, receiving its known value. Neither changes the substance of any result.

4 Two evaluations: flee, explore, or stay

Given only $X_0 = b$, a candidate second location at distance t has $X_t \sim N\big(b\lambda,\, 1-\lambda^2\big)$ with $\lambda = e^{-\kappa t} \in (0,1]$. From (2),

$$\log \mathbb{E}\big[e^{X_t} \,\big|\, X_0 = b\big] \;=\; b\lambda + \tfrac{1}{2}\big(1-\lambda^2\big) \;=\; -\tfrac{1}{2}\,(\lambda - b)^2 + \tfrac{1}{2}\,\big(b^2+1\big)\,, \tag{3}$$

maximized over $\lambda \in [0,1]$ at $\lambda^* = \min\{\max\{b,0\},1\}$.

Proposition 1 (Two-shot policy). The optimal second location and its log-value $\zeta(b)$ are

$$t^* \;=\; \begin{cases} \infty & b<0 \quad \text{(flee)}, \\ -\frac{1}{\kappa}\log b & 0 \le b \le 1 \quad \text{(explore)}, \\ 0 & b>1 \quad \text{(stay)}, \end{cases} \qquad \zeta(b) \;=\; \begin{cases} \frac{1}{2} & b<0, \\ \frac{1}{2}\,\big(b^2+1\big) & 0 \le b \le 1, \\ b & b>1. \end{cases} \tag{4}$$

A below-median first value is abandoned, and an exceptional one is kept: already at two shots, an observed location can be the optimal final location. Practitioners know this fork as pivot or persevere (Ries 2011); the proposition supplies its thresholds and the size of the step when pivoting wins. In between, the searcher steps to the distance at which the prior pull toward the mean exactly balances the anchor's height. The value ζ serves as the exterior benchmark throughout: by the Markov property, any placement beyond all existing observations faces a fresh copy of this problem anchored at the nearest observation.

5 Bridge moments, and what composes how

Suppose values $X_0 = b$ and $X_{t_1} = d$ are known, with bracket correlation $\rho = e^{-\kappa t_1}$ and rapidity $\Theta = \operatorname{artanh}\rho$. For an interior location let $\lambda_2 = e^{-\kappa t_2}$ and $\lambda = e^{-\kappa(t_1 - t_2)}$ be the correlations to the two ends, $\lambda_2\lambda = \rho$, with rapidities $\theta_2 = \operatorname{artanh}\lambda_2$ and $\theta = \operatorname{artanh}\lambda$. Gaussian conditioning gives the bridge moments

$$\mu \;=\; \frac{\lambda_2\big(1-\lambda^2\big)\,b + \lambda\big(1-\lambda_2^2\big)\,d}{1-\rho^2}, \qquad \nu \;=\; \frac{\big(1-\lambda_2^2\big)\,\big(1-\lambda^2\big)}{1-\rho^2}. \tag{5}$$

For $b = d > 0$ the bridge sags: the mean is below b everywhere strictly inside, pulled toward the process mean, while the variance rises from zero at the ends to a maximum at the middle.

A caution on what is additive. Correlations across consecutive intervals multiply, $\rho(t+s) = \rho(t)\,\rho(s)$, so the additive coordinate for composing intervals is $\kappa t = -\log\rho$, not the rapidity. The rapidity earns its keep through a different identity: for equal anchors $d = b$ the bridge mean is

$$\mu \;=\; b\,\frac{\lambda_2+\lambda}{1+\lambda_2\lambda} \;=\; b\tanh(\theta_2+\theta)\,, \tag{6}$$

which matches the mean of the exterior point at correlation $\tanh(\theta_2+\theta)$ to its anchor. This is a mean-matching device we use next, not a composition law. The bridge variance also takes a hyperbolic product form: substituting $1-\tanh^2 = \operatorname{sech}^2$ into (5),

$$\nu \;=\; \operatorname{sech}(\theta_2+\theta)\operatorname{sech}(\theta_2-\theta)\,, \tag{7}$$

while an exterior point at correlation $\lambda' = \tanh\theta'$ to its anchor has variance $1-\tanh^2\theta' = \operatorname{sech}^2\theta'$.

6 Matched-mean dominance at a constant variance ratio

Theorem 1 (Bridge dominance, equal anchors). Let $d = b$ and let $t_2 \in (0, t_1)$ be an interior location with rapidities (θ_2, θ). The exterior location at composed rapidity $\theta' = \theta_2 + \theta$ has the same conditional mean $b\tanh(\theta_2+\theta)$, and the ratio of conditional variances is the same at every interior point:

$$\frac{\nu_{\text{inside}}}{\nu_{\text{outside}}} \;=\; \frac{\cosh(\theta_2+\theta)}{\cosh(\theta_2-\theta)} \;=\; \frac{1+\tanh\theta_2\tanh\theta}{1-\tanh\theta_2\tanh\theta} \;=\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta} \;>\; 1. \tag{8}$$

Consequently the interior point's conditional distribution dominates its mean-matched exterior rival in the Gaussian convex order, and its expected payoff is at least as large for every convex payoff with finite expectation, strictly so for the exponential.

Proof. The mean statement is (6). The ratio of (7) to $\operatorname{sech}^2(\theta_2+\theta)$ is $\cosh(\theta_2+\theta)/\cosh(\theta_2-\theta)$, and expanding both via $\cosh(A\pm B) = \cosh A\cosh B \pm \sinh A\sinh B$ gives $(1+\tanh\theta_2\tanh\theta)/(1-\tanh\theta_2\tanh\theta)$, which is $(1+\rho)/(1-\rho)$ because $\tanh\theta_2\tanh\theta = \lambda_2\lambda = \rho$ on the bracket. Two Gaussians with equal means and ordered variances are ordered in the convex order (the wider equals the narrower plus independent noise), and (2) is the exponential case. □

The amplification is a property of the bracket, not the position: threefold at $\rho = \frac{1}{2}$, nineteenfold at $\rho = 0.9$, unbounded as $\rho \uparrow 1$. At the bracket's ends both variances vanish; their ratio still tends to $e^{2\Theta}$. Dominance over mean-matched exterior rivals, which the theorem explicitly constructs, decides nothing by itself: the exterior player may choose any correlation, and an observed anchor may simply be kept. The object that can fail to exist is different: an *unvisited* point whose conditional mean matches the best *observed* value. Mean reversion caps unvisited conditional means strictly below a high anchor, and whether the variance bonus makes up the shortfall is precisely the question the phase diagram answers. The next section makes this exact.

7 The constrained parabola and the complete symmetric solution

Parametrize the interior by $x = \lambda_2 \in (\rho, 1)$ and set

$$C \;=\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta}, \qquad m \;=\; \frac{x+\rho/x}{1+\rho} \;\in\; \left[\frac{2\sqrt{\rho}}{1+\rho},\, 1\right), \tag{9}$$

under which mirror points $x \leftrightarrow \rho/x$ coincide, the lower endpoint of m is the bracket middle, and, using $\left(1-x^2\right)\left(1-\rho^2/x^2\right) = (1+\rho)^2\left(1-m^2\right)$, the equal-anchor interior log-utility becomes a constrained parabola:

$$L_{\text{inside}}(m) \;=\; b\,m + \frac{C}{2}\left(1-m^2\right), \qquad m^* \;=\; \operatorname{clip}\!\left(\frac{b}{C};\; \frac{2\sqrt{\rho}}{1+\rho},\, 1\right). \tag{10}$$

This is the organizing object of the paper: the entire symmetric analysis reads off the position of the unconstrained peak b/C relative to the interval.

Theorem 2 (Pitchfork). The bracket middle is the interior optimum iff $b/C \le 2\sqrt{\rho}/(1+\rho)$, i.e.

$$b \;\le\; \frac{2\sqrt{\rho}}{1-\rho} \;=\; \sinh 2\tau, \qquad \tau = \operatorname{artanh}\sqrt{\rho}. \tag{11}$$

For $\sinh 2\tau < b < e^{2\Theta}$ the peak is interior and attained by the mirror pair

$$x_{\pm} \;=\; \frac{b\,(1-\rho) \pm \sqrt{b^2\,(1-\rho)^2 - 4\rho}}{2}, \qquad x_+ x_- = \rho, \tag{12}$$

with optimal value

$$U_{\text{off}} \;=\; \frac{C}{2} + \frac{b^2}{2C} \;=\; \tfrac{1}{2}\left(b^2 e^{-2\Theta} + e^{2\Theta}\right). \tag{13}$$

For $b \ge e^{2\Theta}$ the peak $b/C \ge 1$ lies at or beyond the interval's open end: L_{inside} is increasing in m, no interior maximum is attained, and the interior supremum is the end limit b.

Proof. $dL/dm = b - Cm$ vanishes at $m = b/C$; the interval endpoints translate via (9) into the stated b-thresholds ($b/C = 2\sqrt{\rho}/(1+\rho)$ is $b = 2\sqrt{\rho}/(1-\rho)$, and $b/C = 1$ is $b = (1+\rho)/(1-\rho)$; the hyperbolic forms follow from $\tanh\tau = \sqrt{\rho}$). Solving $x + \rho/x = (1+\rho)\,b/C = b\,(1-\rho)$ gives (12), and substituting $m = b/C$ into (10) gives (13). ∎

Theorem 3 (Symmetric phase diagram). For $d = b \geq 0$ the best interior point strictly beats every alternative — exterior placements and observed anchors alike — precisely for

$$b_-(\rho) \ < \ b \ < \ e^{2\Theta}, \qquad b_-(\rho) = \frac{\sqrt{\rho}\left(2 - \sqrt{2\,(1-\rho)}\right)}{1+\rho}, \tag{14}$$

the lower edge being the smaller root of $b^2\,(1+\rho) - 4b\sqrt{\rho} + 2\rho = 0$. For $b \geq e^{2\Theta}$ the optimal final location is an *observed* one: staying at an anchor beats every unvisited point, interior or exterior.

Proof. If $b \geq e^{2\Theta}$, Theorem 2 gives interior supremum b, unattained, while every exterior point loses to the anchor pointwise: for $b > 1$ and any $\lambda \in [0, 1)$,

$$b - \left[b\lambda + \tfrac{1}{2}\left(1 - \lambda^2\right)\right] \ = \ (1-\lambda)\left[b - \tfrac{1+\lambda}{2}\right] \ > \ 0 \tag{15}$$

(the suprema coincide at b; the dominance is strict at every unvisited point): staying at an anchor wins. If $\sinh 2\tau < b < e^{2\Theta}$ the interior optimum is U_{off}, and both comparisons are one line: for $b > 1$,

$$U_{\text{off}} - b \ = \ \tfrac{1}{2}\, e^{-2\Theta}\left(b - e^{2\Theta}\right)^2 \ > \ 0, \tag{16}$$

the arithmetic–geometric mean inequality applied to the two terms of (13), with equality only at the tangency $b = e^{2\Theta}$ defining the upper edge; for $b \leq 1$,

$$U_{\text{off}} - \tfrac{1}{2}\left(b^2 + 1\right) \ = \ \tfrac{1}{2}\left(e^{2\Theta} - 1\right)\left(1 - b^2 e^{-2\Theta}\right) \ > \ 0. \tag{17}$$

If $b \leq \sinh 2\tau$ the optimum is the middle, with value $U_{\text{mid}} = b\,\frac{2\sqrt{\rho}}{1+\rho} + \frac{1-\rho}{2(1+\rho)}$; for $b \leq 1$, $U_{\text{mid}} > \zeta(b)$ is the stated quadratic, and $b_- \leq 2\sqrt{\rho}/\left(1+\rho\right) < \sinh 2\tau$ places the lower edge inside this regime. No gap opens above the middle regime: when $\sinh 2\tau \leq 1$ (i.e. $\rho \leq 3 - 2\sqrt{2}$), the larger quadratic root satisfies $b_+ \geq \sinh 2\tau$, equivalent to $(1-\rho)^3 \geq 8\rho^2$, which holds on that range (left side decreasing, right increasing, valid at the endpoint); when $\sinh 2\tau > 1$, $b_+ \geq 1$ reduces, with $u = \sqrt{\rho}$, to $2u^2\,(1+u) \geq (1-u)^3$, true at $u = \sqrt{2} - 1$ (it reads $20\sqrt{2} \geq 28$) and increasingly so; and on $1 < b \leq \sinh 2\tau$, $U_{\text{mid}} \geq b$ reduces to $b \leq (1+u)\,/\,(2\,(1-u))$, which covers the stretch because $\sinh 2\tau \leq (1+u)\,/\,(2\,(1-u))$ is $(1-u)^2 \geq 0$. $\square$

Along the two-shot-inherited bracket $\rho = b$ the region opens at $b^* = 0.4233\ldots$, the root of $U_{\text{mid}}(b, b) = \zeta(b)$. The geometry of the upper edge is easy to misread: for $b > 1$, interior dominance requires $\rho > (b-1)\,/\,(b+1)$, so higher anchors demand a larger bracket correlation, which is a *shorter* physical bracket. Very good news is forgiven only by narrow brackets; on a wide bracket, very good news means stay.

8 Beyond equal anchors

For general (b, d) the interior stationary points solve the quartic

$$x^4 - (b - \rho d)\,x^3 + \rho\,(d - \rho b)\,x - \rho^2 \ = \ 0, \tag{18}$$

obtained by differentiating (5); its real roots in $(\rho, 1)$, compared against $\zeta(b)$ and $\zeta(d)$, decide the move exactly and replace any grid. The symmetric window does not extend indefinitely in

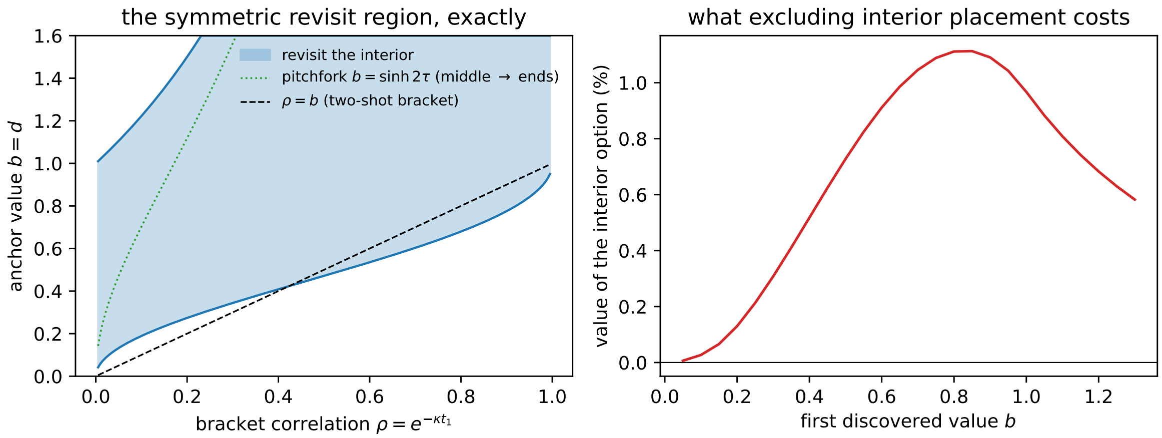

Figure 2: Left: the exact symmetric revisit region (14) in the plane of bracket correlation ρ and anchor value $b = d$, bounded below by the quadratic root b_- and above by $e^{2\Theta}$. The dotted curve is the pitchfork $b = \sinh 2\tau$: below it the optimal interior point is the bracket middle, above it the mirror pair (12) hugging the ends. The dashed line is the two-shot bracket $\rho = b$, entering the region at $b^* = 0.4233$. Right: the value of the interior option — the percentage increase in expected payoff of the full three-shot optimum over the best policy denied interior placements (it may still place on either exterior ray) — as a function of the first discovered value b, with the bracket optimized in both cases.

d. At $b = \frac{1}{2}$, $\rho = \frac{1}{2}$, the interior wins precisely for $d \in (\approx 0.4715, 1)$: below, the ray past the better end wins; at $d = 1$ every strictly interior point loses, by the identity

$$L(x) - 1 \;=\; -\frac{(2x-1)^2\left(x^2+x+1\right)}{6x^2} \;<\; 0, \qquad \tfrac{1}{2} < x < 1, \tag{19}$$

and for $d \geq 1$ staying at the second anchor is optimal, since d's coefficient in the bridge mean is below one and raising d only widens the loss. Symmetric dominance is a window, not a half-line.

9 The three-shot policy, computed

Writing $V(b, d, \rho)$ for the best of $\zeta(b)$, $\zeta(d)$ and the interior optimum (via (18)), the three-shot value of a bracket ρ is $\log \mathbb{E}_d\left[e^{V(b,d,\rho)}\right]$ with $d \sim N\left(b\rho, 1-\rho^2\right)$, computed by adaptive quadrature and optimized over ρ. The comparison policy denies interior placement but keeps both exterior rays. It is not a ban on reversing direction.

Three observations, all in Table 1 and Figure 2. First, a below-median start still means flight, and the interior option is then worthless. Second, the three-shot bracket is wider than the two-shot step, but most of the widening is *not* bought by the interior option: at $b = 0.5$ the two-shot policy steps to $\rho = 0.5$, excluding interior placement already widens the optimum to $\rho = 0.323$, and admitting it widens further to $\rho = 0.289$. The extra evaluation and the choice it enables account for most of the stretch; the interior option adds a further, smaller pull. Third, the interior option's worth peaks near 1.1% of expected value around $b \approx 0.83$

b	optimal ρ	value	value, interior excluded	interior worth
0.1	0.053	0.8396	0.8393	0.03%
0.3	0.167	0.8723	0.8692	0.31%
0.5	0.289	0.9386	0.9314	0.73%
0.7	0.402	1.0375	1.0271	1.05%
0.83	0.469	1.1183	1.1072	1.12%
0.9	0.502	1.1670	1.1562	1.09%
1.2	0.622	1.4059	1.3991	0.68%

Table 1: The three-shot game: optimal bracket correlation, log-value, the log-value with interior placement excluded (exterior rays in both directions permitted), and the percentage the interior option adds to expected payoff. Adaptive quadrature over the second value d; interior optima exact via the stationarity quartic (18). For $b < 0$ the optimal bracket correlation tends to zero and the limiting log-value is exactly $\frac{1}{2} + \log\bigl(1 + 1/\sqrt{2\pi}\bigr) = 0.8357\ldots$

and vanishes at both extremes. Excluding interior placement is thus cheap by value; it is the prescription that errs, and for equal anchors it errs exactly on the revisit window (14) — high, but below the ceiling $e^{2\Theta}$ above which both policies stay. For unequal observations the erring region is a joint condition on (b, d, ρ), decided by the quartic of Section 6, not membership of each value separately: at $\rho = \frac{1}{2}$, $b = \frac{1}{2}$, $d = 2$, both values lie in the symmetric window, yet staying at the second anchor is optimal and the exclusion costs nothing.

10 The landscape premise on real measurements

Whether real search fields look like exponentiated OU paths is measurable. We use the WiFi positioning dataset of Feng, Nguyen and Luo (2024): a university floor of 92×15 meters on a 0.6-meter grid, 642 reference points, 13 access points, 120 repeated measurements per point over three days, released CC BY. Received power is a search field in the paper's literal sense: a deployment seeks the location where signal is strongest, and the landscape is rugged for physical reasons.

For each location and access point we form the mean linear power over the repeats and take its logarithm, then remove an AP-specific log-distance path-loss trend with the access-point position fitted. The 4,235 surviving location–AP pairs form 50 corridor traces of fifteen or more consecutive grid points. The residuals are close to Gaussian (skewness 0.008, excess kurtosis 0.05), and their pooled spatial correlation along corridors is exponential with length 6.6 meters and $R^2 = 0.98$ (Figure 3), the form proposed for shadow fading by Gudmundson (1991) and standard in radio engineering since. Roughly half the variance is a short-scale component below the grid resolution: the surface is the paper's curve plus fine gravel, and the replay below scores against the surface as measured.

A trace-split replay makes the regime point concrete. The model is fitted on half the traces; on each held-out trace a random incumbent's value is revealed, one further measurement is placed by each policy, and a deployment is committed and scored at the measured surface. With correlation length near eleven grid units on traces of fifteen to ninety-two, far probes are nearly fresh draws, and the leading policies are the far-probing ones: probing the far end gains 70% of linear payoff over incumbent reuse and an empirical knowledge gradient 63%, against 33% for

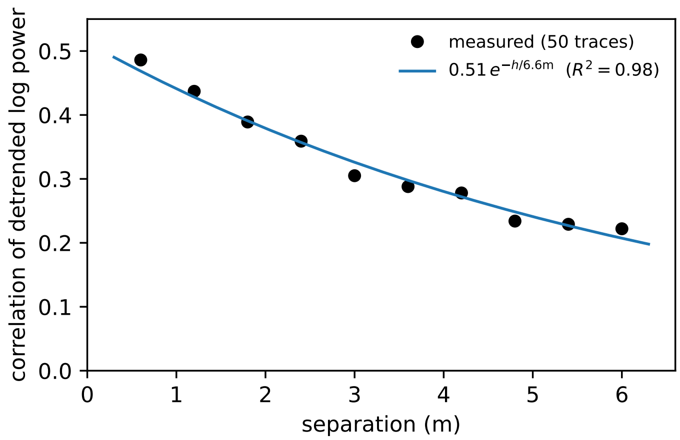

Figure 3: Pooled correlation of detrended log mean power along 50 corridor traces of the Feng et al. dataset, with the fitted exponential. The landscape premise, measured.

the three-shot rule probing at the two-shot step. The exact formulas predict this ordering: at bracket correlation ρ the interior option's premium is $\frac{1}{2}(e^{2\Theta} - 1)(1 - b^2 e^{-2\Theta}) \approx \rho(1 - b^2)$, first order in ρ, while the fresh-draw variance bonus is order one. The revisit machinery governs the opposite regime, search at steps within a correlation length; adjacent grid points here sit at $\rho = 0.91$. Where a firm's feasible pivots are local, the phase diagram is the operative map; where they are global, measure far and take the best. The replay is a first pass: conditioning the comparison on the phase-diagram region of each episode, and per-trace intercepts in the fitted model, are the natural refinements.

11 Discussion

The proverb, read as mathematics, is conditional, and two distinct comparisons should not be merged. Against a *visited* point of matching conditional mean, any unvisited location wins by Jensen alone: it carries variance, the known value carries none. Between two *unvisited* rivals of matching mean, an interior point and its constructed exterior counterpart, Theorem 1 gives the constant variance ratio $e^{2\Theta}$. Neither comparison manufactures a matched-mean unvisited rival for a high observed value. Mean reversion caps the conditional mean of every unvisited point, and above $b = e^{2\Theta}$ the cap binds: the optimal final location is one already seen, and the grass, there, is provably not greener. Below the cap the analysis locates the greenest patch: past the better end on bad news, in whichever direction that end lies; between the anchors on good news, at the middle for moderate anchors and hugging an end beyond the pitchfork.

A caution about the word exploration. With one evaluation remaining there is nothing to explore: no discovery at the final point can inform a later choice. The last move is pure placement, and the pull toward uncertain locations is Jensen's inequality, not curiosity. Exploration proper enters only with more shots remaining. The natural conjecture from this endgame is a two-phase k-shot policy: step out past the running best while values climb, then refine between the two best once a good region is bracketed, with the pitchfork inherited by the refinement phase. There is no irreversible switch, since an adverse interior observation can make leaving attractive again. The bridge formulas make each policy evaluation cheap; the search over adaptive policies that a k-shot dynamic program requires remains to be analyzed, and we make no claim that it is small.

The problem solved here is terminal placement: the payoff is the value at the final location, not the running maximum of the path. Section 2 locates it in the literature; the short version is that the tradition's exact objects are thresholds in $\sigma^2/|\mu|$ on drifting Brownian paths with endogenous evaluations and flow payoffs, and the fixed-budget, terminal-payoff, mean-reverting combination, with its phase diagram, constant amplification $e^{2\Theta}$, and constrained parabola (10), is new.

Reproducibility

All numbers and the figure are produced by `numerics2.py` alongside this source; the review-driven identities (the constant variance ratio, the $d = 1$ factorization, the stationarity quartic, the exact flee limit) are independently checked in `verify/check_review.py`, and the earlier counterexample machinery lives in `verify/check_inside.py` and `verify/check_middle.py`, all in the browniansearch repository.

References

[1] S. Callander. Searching for good policies. *American Political Science Review*, 105(4):643–662, 2011.

[2] S. Callander. Searching and learning by trial and error. *American Economic Review*, 101(6):2277–2308, 2011.

[3] A. Bardhi and S. Callander. Learning in a correlated world. *Annual Review of Economics*, forthcoming.

[4] A. Bardhi. Attributes: selective learning and influence. *Econometrica*, 92(2):311–353, 2024.

[5] C. Urgun and L. Yariv. Contiguous search: exploration and ambition on uncharted terrain. *Journal of Political Economy*, 133, 2025.

[6] Y. F. Wong. Forward-looking experimentation of correlated alternatives. *Theoretical Economics*, 20, 2025.

[7] S. Callander. A theory of policy expertise. *Quarterly Journal of Political Science*, 3(2):123–140, 2008.

[8] S. Callander and P. Hummel. Preemptive policy experimentation. *Econometrica*, 82(4):1509–1528, 2014.

[9] S. Callander and T. S. Clark. Precedent and doctrine in a complicated world. *American Political Science Review*, 111(1):184–203, 2017.

[10] S. Callander and N. Matouschek. The risk of failure: trial and error learning and long-run performance. *American Economic Journal: Microeconomics*, 11(1):44–78, 2019.

[11] S. Callander and N. McCarty. Agenda control under policy uncertainty. *American Journal of Political Science*, 68(1):210–226, 2024.

[12] S. Callander, N. S. Lambert, and N. Matouschek. Innovation and competition on a rugged technological landscape. *American Economic Journal: Microeconomics*, forthcoming.

[13] Y. C. Aybas and S. Callander. Cheap talk in complex environments. Working paper, 2025.

[14] S. C. Ganz. Hyperopic search: organizations learning about managers learning about strategies. *Organization Science*, 31(4):821–838, 2020.

[15] U. Garfagnini and B. Strulovici. Social experimentation with interdependent and expanding technologies. *Review of Economic Studies*, 83(4):1579–1613, 2016.

[16] D. Cetemen, C. Urgun, and L. Yariv. Collective progress: dynamics of exit waves. *Journal of Political Economy*, 131(9):2402–2450, 2023.

[17] C. Carnehl and J. Schneider. A quest for knowledge. *Econometrica*, 93(2):623–659, 2025.

[18] C. Hodgson and G. Lewis. You can lead a horse to water: spatial learning and path dependence in consumer search. *Econometrica*, 93(4):1299–1332, 2025.

[19] D. Glick and C. D. Myers. Learning from others: an experimental test of Brownian motion uncertainty models. *Journal of Theoretical Politics*, 27(4):588–612, 2015.

[20] S. Malladi. Searching in the dark and learning where to look. Working paper, 2025.

[21] M. Banchio and S. Malladi. Rediscovery. Working paper, arXiv:2504.19761, 2025.

[22] J. Grosse, C. Zhang, and P. Hennig. Optimistic optimization of Gaussian process samples. *Transactions on Machine Learning Research*, 2023.

[23] X. Feng, K. A. Nguyen, and Z. Luo. WiFi RTT RSS dataset for indoor positioning. Zenodo, 2024. doi:10.5281/zenodo.11558192.

[24] M. Gudmundson. Correlation model for shadow fading in mobile radio systems. *Electronics Letters*, 27(23):2145–2146, 1991.

[25] E. Ries. *The Lean Startup*. Crown Business, New York, 2011.

[26] A. Camuffo, A. Cordova, A. Gambardella, and C. Spina. A scientific approach to entrepreneurial decision making: evidence from a randomized control trial. *Management Science*, 66(2):564–586, 2020.

[27] A. Camuffo, A. Gambardella, D. Messinese, E. Novelli, E. Paolucci, and C. Spina. A scientific approach to entrepreneurial decision making: large scale replication and extension. *Strategic Management Journal*, 45(6):1209–1237, 2024.

[28] O. Baumann, J. Schmidt, and N. Stieglitz. Effective search in rugged performance landscapes: a review and outlook. *Journal of Management*, 45(1):285–318, 2019.

[29] J.-B. Grill, M. Valko, and R. Munos. Optimistic optimization of a Brownian. In *Advances in Neural Information Processing Systems 31*, 2018.

[30] J. M. Calvin. Average performance of a class of adaptive algorithms for global optimization. *Annals of Applied Probability*, 7(3):711–730, 1997.

[31] A. Alabert and R. Caballero. On the minimum of a conditioned Brownian bridge. *Stochastic Models*, 34(3):269–291, 2018.

[32] R. Agrawal. The continuum-armed bandit problem. *SIAM Journal on Control and Optimization*, 33(6):1926–1951, 1995.